

High-quality vegetation index product generation: A review of NDVI time series reconstruction techniques

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ABSTRACT

Normalized difference vegetation index (NDVI) derived from satellites has been ubiquitously utilized in the field of remote sensing. Nevertheless, there are multitudinous contaminations in NDVI time series because of the atmospheric disturbance, cloud cover, sensor failure, and so on. It is crucial to remove the noises prior to further applications. Numerous techniques have been proposed to alleviate this issue in the last few decades. To the best of our knowledge, there hasn't been a systematical study to summarize and analyze the status of NDVI time series reconstruction techniques since 1980s. As a result, our goal is to recapitulate the current approaches for reconstructing high-quality NDVI time series, followed by an interpretation on the principle, merits and demerits of different kinds of methods. They were mainly classified into temporal-based methods, frequency-based methods and hybrid methods. The evaluation approaches on the quality of NDVI reconstruction were introduced, accompanied with the future development tendency.

1. Introduction

Time-series normalized difference vegetation index (NDVI), derived from remote sensing images, is widely applied in various fields including vegetation change monitoring (Chu et al., 2019; Chuai et al., 2013; Fensholt et al., 2012; Mao et al., 2012; Piao et al., 2011b), landcover type classification (DeFries et al., 1995; Defries and Townshend 1994; Evans and Geerken 2006; Gopal et al., 1999; Hopfner and Scherer 2011), environmental dynamic simulation (Chen et al., 2006b; Jiang et al., 2017; Pettorelli et al., 2011; Tong et al., 2017; Zhao et al., 2020), vegetation phenology extraction (Buyantuyev and Wu 2012; de Jong et al., 2011; Jeong et al., 2011; Pastor-Guzman et al., 2018; Piao et al., 2011a) and so on. As one of the best indicators of vegetation growth state and coverage, NDVI is inversely by the ratio of near-infrared (NIR) and red (R) reflectance as in Eq. (1) (Huete et al., 2002). These two bands, containing over 90% of vegetation information, are ubiquitous in earth observation satellites (Baret et al., 1989).

$$NDVI = (NIR - R) / (NIR + R) \quad (1)$$

With the accumulation of satellite images for the last few decades, there are several kinds of NDVI time series datasets with multiple spatio-

temporal resolutions, mainly derived from Advanced Very High-Resolution Radiometer (AVHRR), SPOT/VEGETATION, Visible Infrared Imaging Radiometer Suite (VIIRS), Moderate Resolution Imaging Spectroradiometer (MODIS), Landsat and Sentinel. Their specific indicators are listed in Table 1. However, they are typically contaminated due to the disturbance of atmosphere conditions, cloud cover, and sensor capacity (Bradley et al., 2007; Kong et al., 2019; Li et al., 2019; Shen et al., 2015; Zhu et al., 2012). Although most of these products have been preprocessed by maximum value compositing (MVC) (Holben 1986), the residual deficiency still remains (Geng et al., 2016; Julien and Sobrino 2019; Wang et al., 2014; Xiao et al., 2017; Zhou et al., 2016a). Therefore, it is indispensable to smooth NDVI time series anterior to the applications.

In general, spatial-based methods are the most common techniques to reconstruct single remote sensing image. Based on the correlation of neighbourhood pixels, they restore the missing pixels according to the valid parts of the image itself. Spatial interpolation is the main representatives, including linear interpolation (Xu et al., 2015), bilinear interpolation (Smith 1981) and kriging interpolation (Bhattacharjee et al., 2014; Zhang et al., 2007). The spatial-based methods may have a good performance for a small number of missing pixels or simple textures. However, they are not appropriate for reconstructing time series

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Table 1
Different NDVI time series products.

Sensors	Products	Resolutions	Dates	Data Sources
MODIS Terra/Aqua	MOD13Q1	16-day, 250 m	1999~	https://ladsweb.modaps.eosdis.nasa.gov
	MOD13A1	16-day, 500 m		
	MOD13A2	16-day, 1 km		
	MOD13A3	Monthly, 1 km		
	MOD13C1	16-day, 0.05 deg		
	MOD13C2	Monthly, 0.05 deg		
VIIRS	VNP13A1	16-day, 500 m	2011~	https://ladsweb.modaps.eosdis.nasa.gov
	VNP13A2	16-day, 1 km		
	VNP13A3	Monthly, 1 km		
	VNP13C1	16-day, 1 km		
	VNP13C2	Monthly, 1 km		
Landsat 5 TM	Landsat5 C1	16-day, 30 m	1984 ~ 2012	https://earthexplorer.usgs.gov
Landsat 7 ETM+	Landsat7 C1	16-day, 30 m	1999~	
Landsat 8 OLI	Landsat8 C1	16-day, 30 m	2013~	
Sentinel-2	S2 L1C	5-day, 10 m	2015~	https://earthexplorer.usgs.gov/
NOAA-AVHRR	GVI	7-day, >=15 km	1982~	https://ladsweb.modaps.eosdis.nasa.gov
SPOT	GIMMS	15-day, 8 km	1981 ~ 2006	https://ecocast.arc.nasa.gov/data/pub/gimms
	VGT-S10	10-day, 1 km	1998~	https://earth.esa.int/eogateway

data with large deficiency. Hence, the spatial-based methods will not be considered in this review.

In the past few decades, dozens of techniques have been exploited to smooth NDVI time series (Geng et al., 2014). According to the principle of missing information restoration, they can be mainly divided into three categories as shown in Fig. 1: 1) temporal-based methods, which are the most prevalent, including four branches: a) temporal interpolation-replacement; b) temporal filter; c) temporal function-fitting methods; d) temporal deep learning models; 2) frequency-based methods; 3) hybrid methods. This classification system is not perfect due to some methods are based on more than one idea. In this case, the uncertain methods will be divided into only one category according to the main ideas behind them. In order to make it more intelligible, their advantages and disadvantages as well as the scope of application are summarized in Table 2.

Moreover, this review will not only summarize the research status of NDVI time series reconstruction (1986–2021) for the first time, but also indicate the potential development tendencies. The ultimate intention is to provide some reference for the related researches and subsequent applications about NDVI time series. The rest parts of this review are organized as follows. Section 2 analyzes the related works and describe the study area, data and methods used in the quantitative evaluation. Sections 3–5 introduce the temporal-based methods, frequency-based

methods and hybrid methods in order. The evaluation techniques of reconstruction performance are recommended in Section 6. The development trends are revealed in Section 7, and Section 8 concludes this review.

2. Data and method

In this section, a detailed meta-analysis was used to disclose the status of NDVI time series reconstruction methods in the past few decades. The study area, experimental data and methods used in the quantitative evaluation were described.

2.1. Meta-analysis on NDVI reconstruction

For literature analysis on NDVI reconstruction, a topic search was carried out in Web of Science core collection (<https://www.webofscience.com/>, accessed September 16, 2021) using the keywords “NDVI time series”, “noise” with “reconstruct*”, “denois*”, “gap-fill*”, “restor*”, “reduc*” or “smooth*”. In total, 206 papers with over 12,000 citations were obtained from 1986 to 2021 as shown in Fig. 2 and Table 3, including 159 journal articles and 47 conference papers.

The publications and citations of NDVI restoration researches from 1986 to 2021 were counted topically, as shown in Table 4. Among the

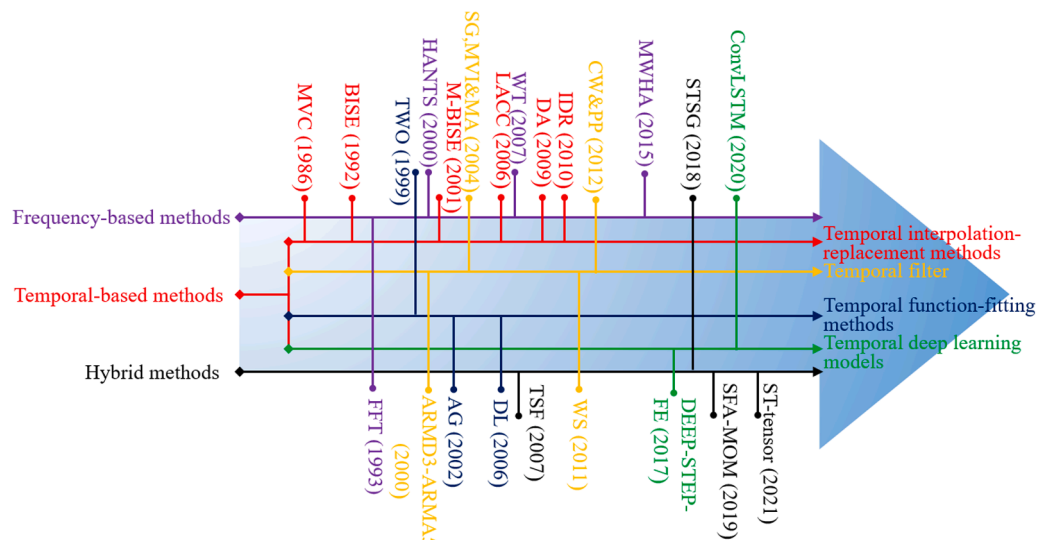
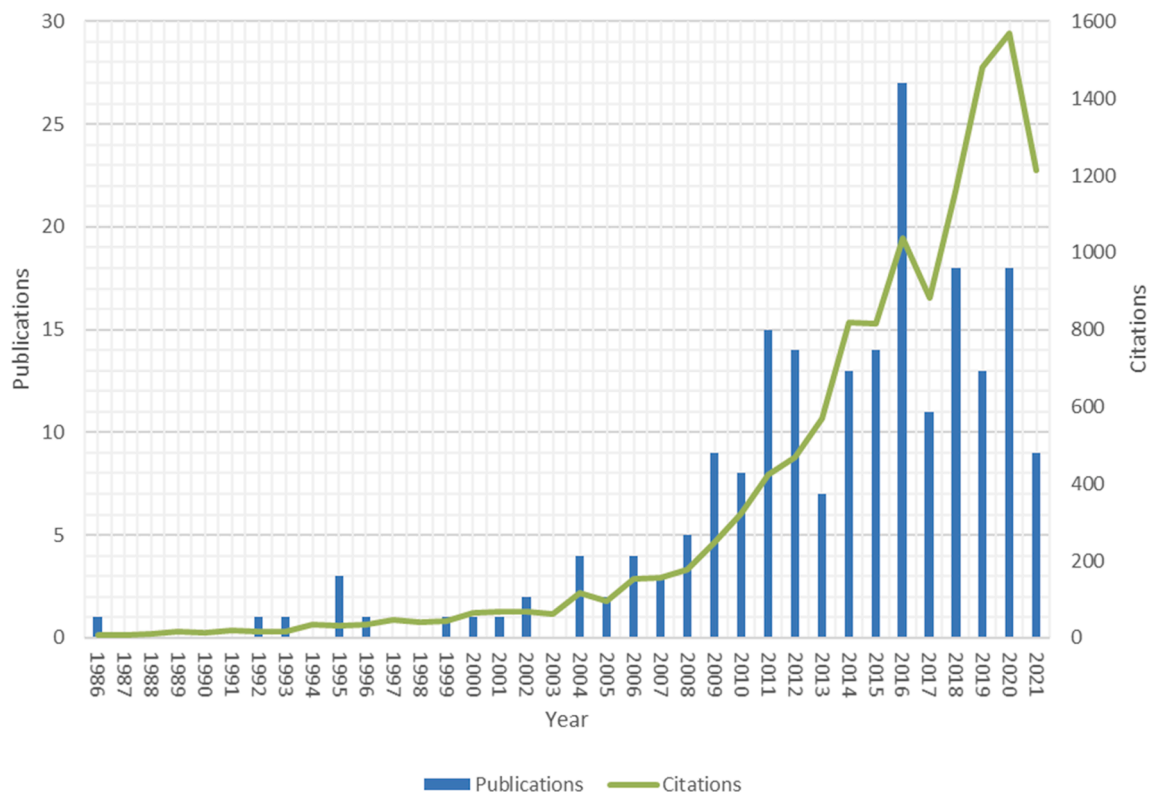


Fig. 1. The development of NDVI time series reconstruction techniques.

Table 2

The advantages, disadvantages and scope of application for each method.

Category	Method	Advantage	Disadvantage	Scope
Temporal interpolation	MVC	Simplicity	Neglecting atmospheric interference	Data without long-term gaps
	IDR	Retaining the valid pixels	Overestimating during dormant period	
	DA	Smoother temporal evolution	Poor performance at the onset of spring green	
Temporal filters	BISE	Close to original time series	Bad for long-term decline trend	Long-term accumulated data
	SG	Reaching the upper envelope	Bad for peak growth and crop harvest	
	MVI	Good with uniform spatial-temporal distribution	Bad for areas with high interannual changes	
	MA	Retaining the seasonal peak of the curve	Changing the width and height of the curve	
Temporal function-fitting	CW	Maintaining the shape and amplitude of the curve	Remaining residual noises	Vegetation without suddenly explodes or perishes
	AG	Extracting phenological parameters while smoothing	Bad for very short growing seasons	
	DL	Revealing the trend of NDVI time series	Remaining problematic with NDVI in winter	
Temporal deep learning models	Deep-STEP_FE	Good performance with a very large dataset	High computational complexity	Nearly all data
Frequency-based methods	FFT	Smooth reconstructed curve	Poor performance for irregular time series	Vegetation without abruptly changes
	HANTS	Reduction of the amount of data	Over-parameterized	
	WT	Preserving patterns of time series while smoothing	Reducing some reasonable high values	
Hybrid methods	TSF	Restoring the seasonal trends	Restriction of QC data	Nearly all data
	SFA-MOM	Good performance in data deficient scenario	Restriction of LULC data	
	STSG	Available for local low values during harvest	Bad for the data with high spatial resolution	
	ST-tensor	Available for spatio-temporal continuous gaps	Restriction of RI data	

**Fig. 2.** Publications and citations about NDVI reconstruction (1986–2021).

three categories, **temporal-based methods are the most popular because the variation of NDVI is time continuous during vegetation growth, which is the most intuitive information that can be utilized for reconstruction.** In the recent decade, hybrid methods using spatio-temporal information are getting more attention for robustness and high performance. As for

study area, regional areas are the most common choice to verify the validity of the reconstruction methods and this is the reason why most methods lack universality. MODIS and AVHRR data are the most used NDVI products, in which MODIS products are widely applied in global evaluation of reconstruction methods. The former is due to moderate

Table 3
Publications on NDVI reconstruction (1986–2021).

Publications/Conferences	Count
Remote Sensing	30
Remote Sensing of Environment	25
International Journal of Remote Sensing	18
IEEE International Geoscience and Remote Sensing Symposium (IGARSS)	12
IEEE Transactions on Geoscience and Remote Sensing	10
ISPRS Journal of Photogrammetry and Remote Sensing	6
International Conference on Agro Geoinformatics	5
International Journal of Applied Earth Observation and Geoinformation	5
Proceedings of SPIE	5
GIScience & Remote Sensing	3
IEEE Geoscience and Remote Sensing Letters	2
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	2

Table 4
Publications and citations for different topics (1986–2021).

Topic	Category	Publications	Citations
Method	Temporal-based	91	8033
	Frequency-based	42	1704
	Hybrid	20	428
Study area	Regional	173	10,138
	Global	19	1580
	Continental	14	1219
Data	MODIS	122	4772
	NOAA-AVHRR	56	8432
	Landsat	36	861
	SPOT	20	1932
	Sentinel	9	73
	VIIRS	3	73

spatio-temporal resolution, while the latter is due to sequences accumulated over a long time.

2.2. Study area and data

In order to evaluate different methods, simulated experiments were conducted using MOD13A1 NDVI products with 500 m resolution synthesized by MVC in 16 days and its corresponding quality control data

from 2016 to 2020. The Shennongjia Forestry District in Hubei, China is selected as the study area, which locates between 109°56'–110°58' E and 31°15'–31°75' as shown in Fig. 3. This region with an area over 3000 km² is covered by clouds severely in summer and autumn. The proportion of invalid pixels is about 40% annually, which is beneficial for verifying each method. 10-year average (2011–2020) of NDVI was chosen as the reference data. According to the quality control data, 5-years (2016–2020) gaps were simulated into the corresponding reference NDVI to generate noise data. After reconstructed, CC and RMSE between the reconstructed data and reference data were calculated. The specific results will be presented in Section 6.2.

3. Temporal-based methods

The temporal-based methods are the most common reconstruction techniques based on the temporal correlation of dataset or phenological characteristics of vegetation. Accordingly, they can be divided into four categories: a) temporal interpolation-replacement methods, like MVC (Holben 1986; Roy 1997; Taddei 1997), iterative interpolation for data reconstruction (IDR) (Julien and Sobrino 2010), data assimilation (DA) (Cai et al., 2011; Gu et al., 2008; Gu et al., 2009), the best index slope extraction (BISE) (Lange et al., 2017; Viovy et al., 1992) and modified BISE (M-BISE) (Lovell and Graetz 2001); b) temporal filter, including Savitzky-Golay (SG) filter (Chen et al., 2004; Li et al., 2020; Tang et al., 2020), mean value iterative (MVI) filter (Ma and Veroustraete 2006), moving average (MA) filter (Jönsson and Eklundh, 2004) and changing-weight (CW) filter (Zhu et al., 2012); c) temporal function-fitting methods, such as asymmetric Gaussian (AG) function fitting (Jonsson and Eklundh 2002; Vorobiova and Chernov 2017), double logistic (DL) function fitting (Beck et al., 2006; Yang et al., 2019) and temporal window operation (TWO) (Park et al., 1999); d) temporal deep learning models, mainly including deep-learning-based framework (Deep-STEP_FE) (Das and Ghosh 2017).

3.1. Temporal interpolation-replacement methods

Among the temporal-based methods, temporal interpolation-replacement is the most basic. They are generally based on certain criteria for the identification of data gaps. As shown in Fig. 4, the conspicuous noises will be replaced by linear or nonlinear temporal

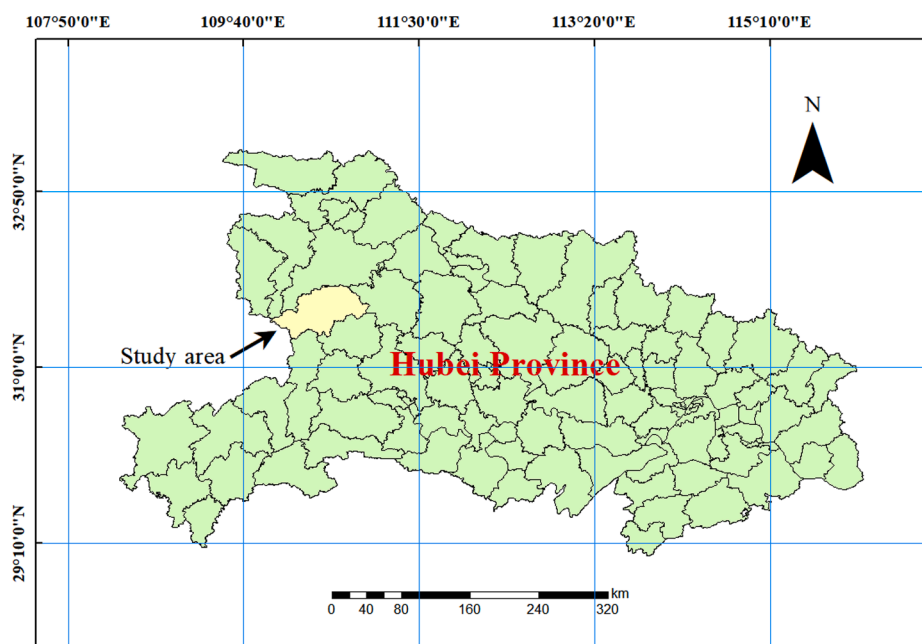


Fig. 3. Location of the study area.

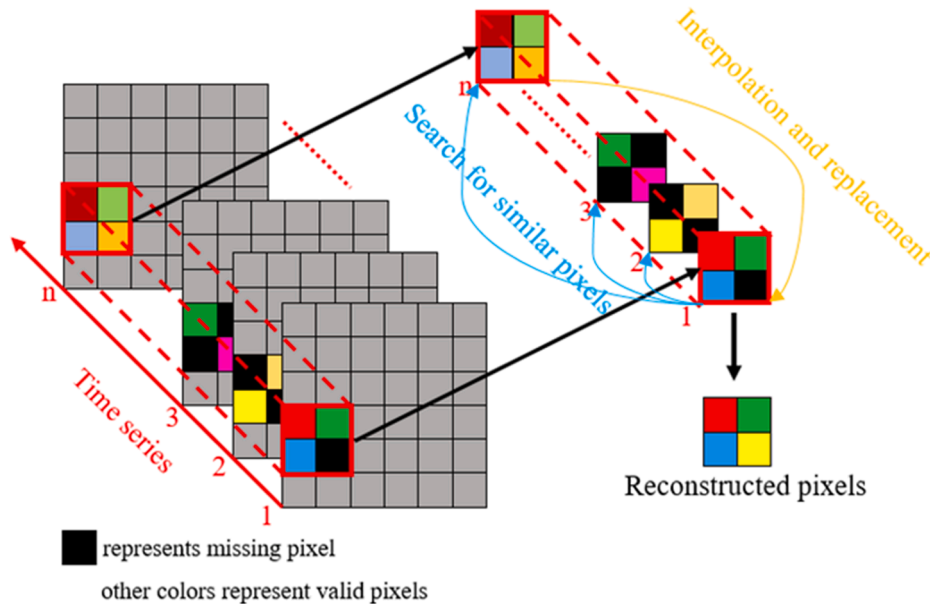


Fig. 4. Temporal interpolation-replacement methods.

interpolation.

3.1.1. Maximum value compositing

As one of the first proposed methods in 1986, MVC is widely-used for NDVI reconstruction. It assumes that the location of clouds is constantly changing and that there are cloud-free days at any location in a certain period (Carreiras et al., 2003). The maximum during the composition period is selected as the reconstructed NDVI by Eq. (2), because the low values are generally regarded as contaminated data (Cao et al., 2018).

$$NDVI_c = \text{Argmax}(NDVI_1, NDVI_2, \dots, NDVI_n) \quad (2)$$

where $NDVI_c$ represents the composited NDVI, n indicates the number of days for composition.

MVC can remove the clouds to some extent, and the best composition period is set to 2–4 weeks in order to reflect the real characteristics of vegetation growth (Viovy et al., 1992). For the data without atmospheric correction, it is still the best preprocessing scheme (Holben 1986). However, it doesn't consider the interference of atmosphere which leads to the loss of useful temporal information.

3.1.2. Iterative interpolation for data reconstruction

IDR is based on the reality that clouds and atmospheric interference tend to reduce NDVI and that the variations of NDVI time series are successive (Julien and Sobrino 2010). The noises in original time series are replaced by the corresponding points in reference data with the largest deviation. This is an iterative process, in which low quality or discontinuous data points increase up to the upper envelope gradually. It shows outperformance in retaining the original valid pixels as well as approaching the upper envelope of NDVI time series. However, it may overestimate NDVI during vegetation dormant period, especially the land surface with frequent variations (Liu et al., 2017).

3.1.3. Data assimilation

DA is simple but fairly effective when there are long-term accumulated data. In brief, the noisy time series are reconstructed by the background data and observed data with the weights in Eq. (3). Moreover, the background data, which can precisely reflect the dynamic change characteristics of vegetation over years, are generated by multi-year NDVI time series based on a three-point smoothing technique.

$$NDVI_r = NDVI_b + w \times (NDVI_o - NDVI_b) \quad (3)$$

where $NDVI_r$ denotes the reconstructed data, $NDVI_o$ represents the observed data, $NDVI_b$ means the background data and w is the weight generated by Eq. (4).

$$w = \frac{B_{NDVI}}{B_{NDVI} + O_{NDVI}} \quad (4)$$

where B_{NDVI} and O_{NDVI} are the background and observation error variances of NDVI respectively.

On the whole, DA is easy to understand and implement. This parameterless method can identify and amend the data gaps efficaciously with QA flag data and show a smoother temporal evolution. Nevertheless, multi-year original data are needed to generate background time series, which may limit its scope of application. In addition, it has bad performance when spring green starts due to the average background (Gu et al., 2009).

3.1.4. The best index slope extraction

Compared with the frequently changed sensor view field and clouds, vegetation variation is more disciplinary and divivable (Viovy et al., 1992). BISE is precisely based on this assumption to reserve high value by an effective detection strategy: searching in the presupposed sliding window, if the value of the latter point is larger than the previous one, the high value is retained; if the value of the latter point is less than the previous one, certain judgment condition is applied to determine whether the point is retained or not. If the value of no point is larger than 20% of the deviation between the low value and the previous high value in this sliding window, the low value will be retained; if the latter point is 0.1 larger than the previous one, the high point will be discarded.

The advantage of BISE is that it doesn't interfere with the slow changing points and the results are usually close to the original NDVI time series (Yang et al., 2015). However, it is extremely sensitive to the size of the sliding window. If the sliding window is too small, much residual noise remains; if it's too large, the time series will be over-smoothing and the detailed information may get obliterated. The window size and threshold are empirical, while a large number of simulation tests are needed to determine the indicators (Lu et al., 2007). This method is originally designed for daily NDVI time series, requiring adjustments for multi-day synthetic data. Lovell and Graetz, 2001 considered the local change rate for improvement, called modified-BISE (M-BISE), which overcomes the drawback that BISE can't process the

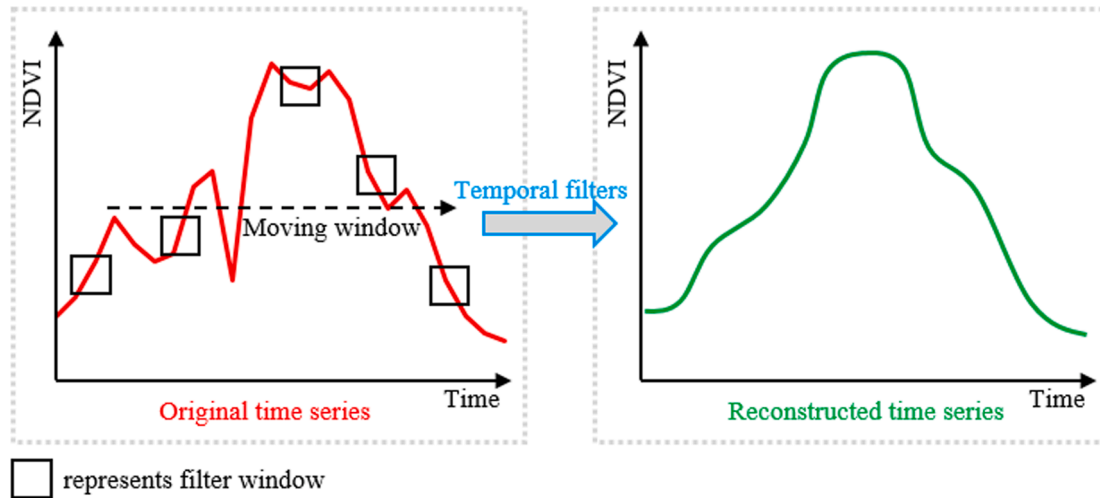


Fig. 5. Temporal filter.

data with a long-term decline trend, and maintains the spatio-temporal variation characteristics of NDVI time series.

In summary, all the temporal interpolation-replacement methods are theoretically simple and undemanding to implement. They have good performance in most cases if the quality of selected time series is high enough. Nevertheless, they are data-sensitive and will produce poor results when there are lots of gaps.

3.2. Temporal filter

The temporal filter methods decompose multi-temporal remote sensing images into one-dimensional time signals and process one pixel each time with an appropriate filter. **They are based on the truth that the fluctuation of NDVI time series is consistent with the phenological curve of vegetation growth (Hird and McDermid 2009).** The brief principle of temporal filter is shown in Fig. 5.

3.2.1. Savitzky-Golay filter

SG filter is a widely used denoising method based on least square fitting (Sa-ing et al., 2018). Chen et al. (2004) firstly introduced it into NDVI time reconstruction. Through calculating the minimum root mean square error (RMSE) in the given sliding window (Sadeghi et al., 2020), it can eliminate the noises deviating from the normal envelope. The reconstructed NDVI can be generated by Eq. (5).

$$NDVI_i^r = \left(\sum_{j=-m}^m C_j \times NDVI_{i+j} \right) / (2m+1) \quad (5)$$

where $NDVI_i$ means the i -th original data, $NDVI^r$ means the reconstructed data, C_j is the coefficient of j -th point in the sliding window, and m is the radius of the sliding window.

According to Eq. (5), SG filter needs to select two indicators manually: one is the radius of the sliding window, and the larger the radius is, the smoother the reconstructed NDVI time series are; the other is the order of the polynomial (generally 2–4), and the lower the order is, the smoother the reconstructed NDVI time series are (Zhao et al., 2014). The detailed procedure of SG filter is shown as Fig. A1 in the Appendix. SG filter is to approach to the envelope of NDVI time series as much as possible rather than simulate the phenological curve of vegetation. Hence, it has a poor performance on excessive cloud contamination scenario or during vegetation growth period (Yang et al., 2015). It requires priori knowledge to determine the width of sliding window and the order of the polynomial, which may limit its availability (Liu et al., 2016). In this case, Tang et al. (2020) improved SG filter with the adaptive optimum-width of sliding window based on the unbiased risk

estimator theory (Stein 1981). Also, some points are added into the original time series to make up a complete sliding window in order to avoid the edge-effect in SG filter (Menon and Seelamantula 2014). This will reduce the accuracy of valid points in the sliding window.

3.2.2. Mean-value iterative filter

MVI filter is a nonlinear smoother based on the reality that the phenology of vegetation is consecutive and gradual. $NDVI_i$ can be substituted with the average of the anterior data $NDVI_{i-1}$ and the subsequent data $NDVI_{i+1}$ if the maximum deviation between $NDVI_i$ and the average is larger than the given threshold. Like the most methods mentioned above, it is also an iterative process. The stopping criterion for iteration (threshold) can be obtained from the multi-year average in practical applications.

MVI is effortless to implement, and it has excellent performance when smoothing data with homogeneous spatio-temporal distribution. However, errors and uncertainties may be introduced due to the substitution and iteration. The selection of threshold will also influence the reconstruction. If the threshold is too high, most of the original values will be preserved; and if it is too low, the NDVI time series will be over fitted (Ma and Veroustraete 2006), which is bad for short-term time series and areas with high interannual changes (Yang et al., 2015).

3.2.3. Moving average filter

MA filter reconstructs NDVI time series by substituting the central point with the average of all the points in a moving window with a fixed size based on Eq. (6).

$$\hat{y} = \sum_{j=-n}^n c_j y_{i+j} \quad (6)$$

where $\hat{y}$ denotes the reconstructed value, $2n+1$ is the size of moving window, c_j represents the weight of each point in the moving window, and y_{i+j} means the value of each point. In general, the closer the point is to the center, the greater the weight is.

MA filter retains the average position and area of the seasonal peak perfectly with low computation complexity. However, this method will change the width and height of the time series due to the choice of weight for each point (Canan et al., 1998).

3.2.4. Changing-weight filter

CW filter proposes two special strategies to process NDVI time series: 1) to determine the local maximum/minimum points in a growth cycle by mathematical morphology algorithm (Soille 2002; Soille 2013) and rules-based decision process; 2) to process the time series using a three-

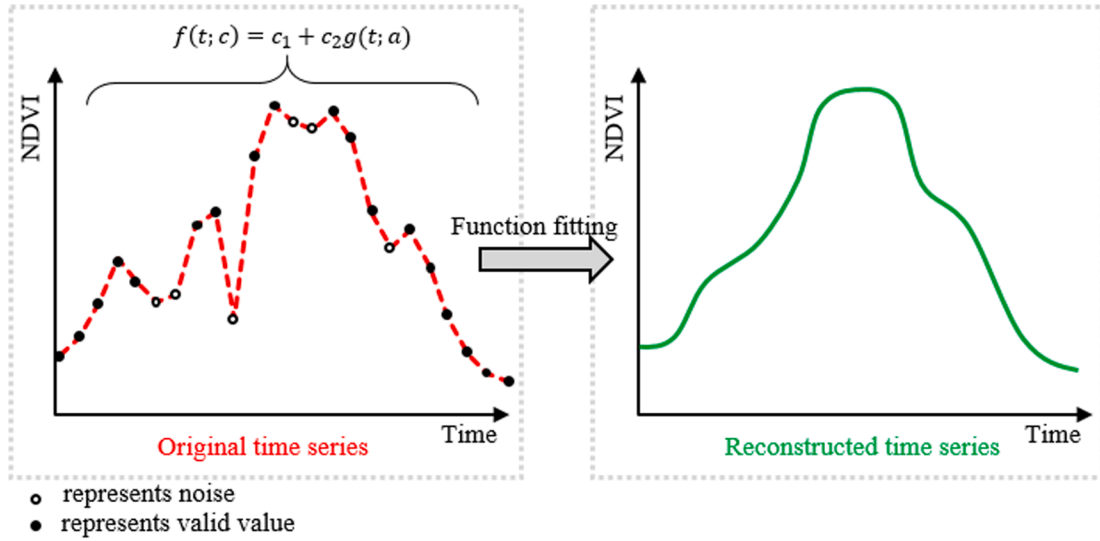


Fig. 6. Temporal function-fitting methods.

point changing-weight convolution filter. The detailed procedure of CW filter is shown as Fig. B1 in the Appendix.

It firstly identifies the local maximum/minimum points in a growth cycle. These points, which are closely related to the phenology of vegetation (Galford et al., 2008), remain constant in the subsequent filtering process to maintain the shape and amplitude of original NDVI time series. In CW, the original NDVI time series are restored by a three-point changing-weight filter based on Eq. (7).

$$\hat{y}_i = \frac{1}{k+2} \cdot y_{i-1} + \frac{k}{k+2} \cdot y_i + \frac{1}{k+2} \cdot y_{i+1}, \quad k = 1, 2, \dots, n \quad (7)$$

where $\hat{y}_i$ denotes the smoothed NDVI value, y_{i-1} , y_i , and y_{i+1} represent the first value, the middle value and the third value in the first filtering window respectively, and k is the iteration number.

During the iteration, the weight of the intermediate value increases while the weight of the two sides decreases. After each iteration, the recognized maximum/minimum position is restored to the original value. Then, the deviation between two adjacent time series will be calculated. If the deviation is less than the given threshold, the corresponding point in the first time series is accepted; if not, the corresponding point in the other one will be adopted to generate a new time series. This iteration will continue until the maximum deviation of the two adjacent time series is less than the given threshold.

By recognizing the true maxima/minima during vegetation growth circle, CW filter can maintain the shape and amplitude of time series. It can also effectively smooth time series while retaining the temporal profile shape (Zhu et al., 2012). Unlike other filter-based methods, the window width of CW is changing according to the shortest length of a vegetation growth cycle, which makes it more flexible to deal with different situations. However, due to the retention of local minima values, CW may preserve noises. Later, CW was improved by using a Gaussian kernel to preserve the phenological (PP) characteristics of vegetation (Jiang et al., 2012).

In addition to the methods mentioned above, there are some other temporal filters such as ARMD3-ARMA5 filter (Filipova-Racheva and Hall-Beyer 2000) and Whittaker smoother (WS) (Atzberger and Eilers 2011; Eilers 2003). In a word, temporal filter methods smooth time series in a moving window according to some certain rules. Although these methods are mainly used to reconstruct NDVI, they can also deal with other time series data with suitable modifications. In general, the reconstructed results of temporal filters meet the phenological curves unless there are plunges or surges, while the restored pixels may be a little different from the corresponding original valid pixels.

3.3. Temporal function-fitting methods

Temporal function-fitting methods construct a specific function to smooth NDVI time series by simulating the vegetation growth rules, mainly including AG and DL. Their fundamental is shown as Fig. 6.

3.3.1. Asymmetric Gaussian function fitting

AG is a nonlinear least square fitting algorithm based on asymmetric Gaussian function as shown in Eq. (8) (Zhao and Zhang 2018).

$$f(t) = f(t; c_1, c_2, a_1, \dots, a_5) = c_1 + c_2 g(t; a_1, \dots, a_5) \quad (8)$$

where $g(t; a_1, \dots, a_5)$ is Gaussian function based on Eq. (9).

$$g(t; a_1, \dots, a_5) = \begin{cases} \exp \left[- \left(\frac{t - a_1}{a_2} \right)^{a_3} \right], & \text{if } t > a_1 \\ \exp \left[- \left(\frac{a_1 - t}{a_4} \right)^{a_5} \right], & \text{if } t < a_1 \end{cases} \quad (9)$$

where c_1 and c_2 control the reference and amplitude of the curve respectively, a_1 determines the position of maximum/minimum, and a_2 , a_3 , a_4 , and a_5 master the width and steepness of the curve.

The weighted least squares fitting process is to obtain the combination of the parameters ($c_1, c_2, a_1, \dots, a_5$) when χ^2 is minimized as shown in Eq. (10).

$$\chi^2 = \sum_{i=n_1}^{n_2} \left[\frac{f(t; c_1, c_2, a_1, \dots, a_5) - I_i}{\sigma^i} \right]^2 \quad (10)$$

where I_i represents the i -th value of original NDVI time series, σ^i denotes the inverse of the i -th weight.

AG shows fairly good performance to adopt the upper envelope under some special circumstances. It is able to describe the seasonal vegetation growth curves and estimate phenological parameters while smoothing over a large-scale region (Jonsson and Eklundh 2002). However, a series of maxima and minima need to be confirmed. It's hard to extract these two parameters when there is much noise in original NDVI time series or the growing season is too short (Lu et al., 2007), which will lead to loss of details about vegetation changes.

3.3.2. Double logistic function fitting

DL tends to regard NDVI time series as a time fitting function with 6 parameters (Yang et al., 2019) with Eq. (11).

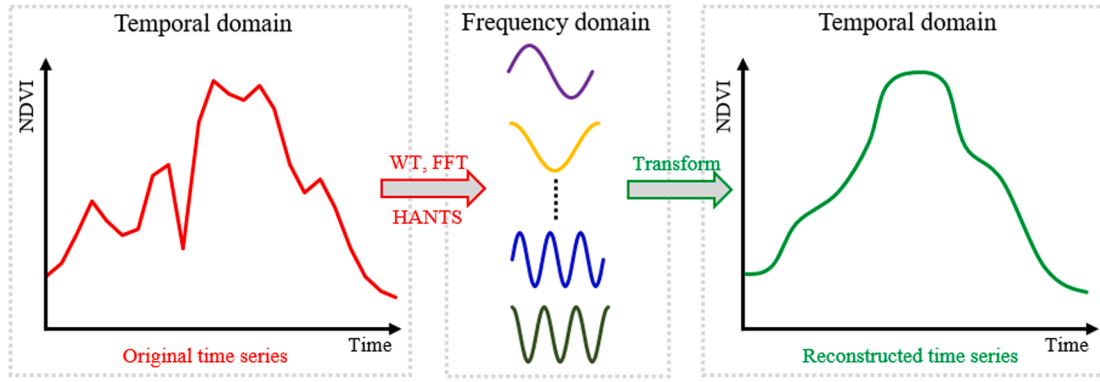


Fig. 7. Frequency-based methods.

$$NDVI = wN + (mN - wN) \times \left(\frac{1}{1 + \exp(-mS \times (t - S))} + \frac{1}{1 + \exp(mA \times (t - A))} - 1 \right) \quad (11)$$

where wN represents winter NDVI, mN represents maximum NDVI, S denotes the position of ascending inflection point, mS denotes the ascending speed, A means the position of declining inflection point, mA means the declining speed and t is the time.

For each pixel point and each year, DL processes original NDVI time series by iterative nonlinear least squares fitting using winter NDVI and other five estimated parameters. Due to the residual error generated during the first fitting, the weights of the observed values need to be set, and the five parameters other than winter NDVI should be reestimated during the second iteration. DL has a good performance when fitting the curve outside the growing season, which can reveal the trend of time series in both space and time. However, too many parameters to be determined will greatly affect the performance when denoising the data of boreal vegetation in winter (Beck et al., 2006).

On the whole, the temporal function-fitting methods are widely used to restore NDVI time series. These methods are committed to make the original curve approximate to the upper envelope. This strategy is quite useful upon most occasions unless vegetation suddenly explodes or perishes.

3.4. Temporal deep learning models

Recently, deep learning has been widely applied in remote sensing field, such as land cover classification (Zhang et al., 2018a; Zhao and Du 2016; Zhao et al., 2015), crop yield forecast (Fernandes et al., 2017; Panda et al., 2010; Pantazi et al., 2016), and land surface temperature (LST) application (Mao et al., 2018; Tan et al., 2019; Wu et al., 2019). However, only few temporal deep learning models have been proposed to reconstruct NDVI time series. Das and Ghosh proposed the Deep-STEP_FE method for reconstructing the missing data with several forecasting modules based on deep stacking network (DSN) (Deng and Yu 2014) as shown in Fig. C1 of the Appendix. Every forecasting module consists of four architectures, including feature set preparation, spatio-temporal feature learning, prediction and dummy image/data tuning (Das and Ghosh 2017) as shown in Fig. D1 of the Appendix. Deep-STEP_FE tends to use the valid historical data from both earlier and subsequent timestamps to obtain missing information, while maintaining the spatio-temporal local constraints.

Generally speaking, most of the temporal-based methods try to find out an explicit relevance between the contaminated and valid data. However, this deep-learning-based technique breaks through the limitation of traditional methods, including parameterized complexity and application limitation, and improves the reconstruction performance effectively. In addition, the powerful computing capacity of temporal deep learning models can help to process a very large dataset.

In summary, temporal-based methods are based on the pollution randomness and phenological regularity of NDVI time series. In most instances, they are able to effectively alleviate the noisy unless there are long-time gaps which may lower the reconstruction precision.

4. Frequency-based methods

The frequency-based methods restore NDVI time series by transforming the contaminated data from temporal domain to frequency domain as shown in Fig. 7. It is believed that the noise is usually the high frequency component which is easy to remove in frequency domain. They mainly include fast Fourier transform (FFT) (Menenti et al., 1993), harmonic analysis of time series (HANTS) (Roerink et al., 2000) and wavelet transform (WT) (Lu et al., 2007; Quiroz et al., 2011).

4.1. Fast Fourier transform

FFT is effective to deal with periodic time series signals. It is based on the hypothesis that a continuous time series signal can be represented as an infinite superposition of sine waves with different frequencies as shown in Eq. (12) (Xu and Shen 2013).

$$F(u) = \sum_{t=0}^{N-1} f(t) e^{-i \frac{2\pi}{N} ut} \quad (12)$$

where $F(u)$ denotes the signal after Fourier transform, $f(t)$ denotes the time series, N is the length of time series, t represents the time domain and u represents the frequency domain.

Firstly, the time series data are transformed into the frequency domain. Since most of the noise belongs to high frequency signals (Bold 1985; Lin and Ye 2019; Liu and Tang 1998), then the transformed data are denoised by low pass filter with Eq. (13).

$$G(u) = F(u) * H(u) \quad (13)$$

where $G(u)$ denotes the signals in frequency domain after filtering and $H(u)$ is an ideal low pass filter as shown in Eq. (14).

$$H(u) = \begin{cases} 1 & D(u) \leq D_0 \\ 0 & D(u) > D_0 \end{cases} \quad (14)$$

where D_0 is the cut-off frequency and $D(u)$ is the absolute distance between the frequency plane point and the origin point.

Finally, the filtered data are processed by the inverse Fourier transform to generate a new noise-free NDVI time series. FFT is simple and its reconstruction curve is very smooth, but it shows poor performance when smoothing irregular time series due to the strict symmetry of the formula (Sellers et al., 1994). Chen et al. (2006a) proposed a locally adjusted cubic-spline capping (LACC) method to address this issue, which is robust to restore any vegetation variation patterns.

Table 5

The parameter description of HANTS algorithm.

Parameter	Description
NoF	Number of frequencies.
FET	Fit error tolerance, the threshold to stop iteration.
Hilo	Direction of outliers ('Hi' represents suppress high value, 'Lo' represents suppress low value and 'none' represents no suppression), and Hilo is usually set to 'Lo' when smoothing NDVI time series.
High	Maximum valid data.
Low	Minimum valid data.
DOD	Degree of over determinedness, the minimum number of samples to solve the parameters.

4.2. Harmonic analysis of time series

Roerink et al. (2000) proposed a classical reconstruction method based on Fourier analysis, abbreviated as HANTS. As one of the most popular denoising techniques, HANTS can smooth the NDVI data using a few important harmonics of the time series (Malamiri et al., 2020). It is an improvement of Fourier method and represents the time series as the combination of several sines and cosines by Eq. (15).

$$y(t) = a_0 + \sum_{i=1}^n [a_i \cos(2\pi f_i t) + b_i \sin(2\pi f_i t)], \quad t = 1, 2, \dots, N \quad (15)$$

where a_0 is the average of NDVI time series, a_i and b_i represent the parameters of sines and cosines, f_i denotes the frequency, t is the number of points in time series, N is the length of time series, and n means the number of harmonic ($2n + 1 \leq N$).

HANTS selects the first several significant frequencies for least square fitting, and the rationality of parameter setting is directly related to the accuracy of reconstruction (Zhou et al., 2016b). In general, six parameters need to be determined empirically as shown in Table 5. The noise points are identified by the effective range of data itself. For NDVI, the high value is set to 1, the low value is set to 0, and the points outside this range are marked as noise. In addition, an iterative least square fitting will be applied during the restoration until the maximum deviation between the newly fitted time series and the input original time series is less than the given threshold (FET) or the remaining points (<DOD) fail to solve the above equation.

HANTS performs well as long as the time series can provide enough valid information to be decomposed to several sine and cosine functions. In other words, the reconstruction results of HANTS may show large deviation with original NDVI time series if there are long-term data gaps (Xu et al., 2015). To overcome this weakness, Yang et al. (2015) proposed the moving weighted harmonic analysis (MWha) method with a partial model using weighted least squares distance. Also, too many parameters need to be determined and this imperfection will definitely reduce the performance and availability of HANTS (Wen et al., 2004).

4.3. Wavelet transform

WT is an effective method to analyze time series in frequency domain (Obidin and Serebrovski 2014). It extracts the features of signals by transformation and localization analysis of time/space frequency (Pan et al., 2008). In this way, it can maintain the useful information while denoising. WT is a group of subfunctions composed of wavelet generating functions after stretching and shifting by Eq. (16).

$$\Psi_{a,b}(x) = \frac{1}{\sqrt{|a|}} \Psi\left(\frac{x-b}{a}\right) \quad (16)$$

where a is the stretching parameter, b is the shifting parameter, $\Psi(x)$ denotes the wavelet generating function and $\Psi_{a,b}(x)$ denotes the function group. The duration of $\Psi(x)$ is fairly limited due to its tight supportive so that wavelet analysis can accurately capture the local characteristics of non-stationary signals. The wavelet of $f(x)$ is shown as

Eq. (17).

$$f_{\text{wavelet}}(a, b) = \int_{-\infty}^{+\infty} f(x) \Psi_{a,b}(x) dx \quad (17)$$

In WT, time series are decomposed at different scales or resolutions and at different displacements. The denoising strategies consist of two parts: 1) the obvious noise is firstly removed by inverse distance weighted interpolation (IDWI) (Bartier and Keller 1996) using the valid information of adjacent temporal images; 2) the residual noise is smoothed by WT.

Unlike other frequency-based methods, WT doesn't depend on any default parameters which makes it fairly robust in most occasions. WT is able to preserve the real patterns of time series change while smoothing (Lu et al., 2007). It is easy to replicate in any software with integrated wavelet functions like MATLAB. However, it may reduce some reasonable high values when there are frequent variations (Yang et al., 2015).

In summary, all the frequency-based methods are easy to implement due to low complexity. These methods depend on an assumption that the high frequency components are usually noise. However, when abrupt changes occur in vegetation growth, such as fire disaster and crop harvest, this kind of variation may also be regarded as noise in frequency domain, and frequency-based methods will fail to maintain vegetation phenology effectively.

5. Hybrid methods

At present, most reconstruction algorithms emphasize temporal continuity while ignore spatial correlation. They adopt a pixel-by-pixel processing approach, which extracts the time series of each pixel prior to smooth. However, this kind of strategy will lead to poor performance on large data gaps. In this case, hybrid methods consider both time continuity and space similarity to overcome the above problems. Four typical hybrid methods using spatio-temporal information will be introduced in the following sections, including temporal spatial (TSF) filter (Fang et al., 2007), search and fill algorithm with moving offset method (SFA-MOM) (Padhee and Dutta 2019), spatio-temporal Savitzky-Golay (STSG) method (Cao et al., 2018) and spatio-temporal tensor completion (ST-Tensor) method (Chu et al., 2021).

5.1. Temporal spatial filter

In TSF, temporal and spatial characteristics of different vegetations are considered to reconstruct MODIS leaf area index (LAI) time series by processing the low quality (QC > 32) and missing pixels (Fang et al., 2008). It needs some appropriate adjustments when recovering NDVI time series. The detailed procedure of TSF is shown as Fig. E1 in the Appendix.

In TSF, the key point is to calculate the background and observation values. For the same type of vegetation, the missing pixels are filled by multi-year average. If the average can't be obtained due to data deficiency, then the background value will be calculated by an ecosystem curve fitting (ECF) algorithm (Moody et al., 2005) using the MODIS vegetation continuous field (VCF) products (Hansen et al., 2003). In this way, the dependence of each target pixel on the phenology of regional continuous field is strengthened, which guarantees the spatio-temporal integrity in pixel level. The observation values are obtained by SG filter. The final reconstructed data α_t is calculated by background value α_b and observed value α_o by Eq. (18).

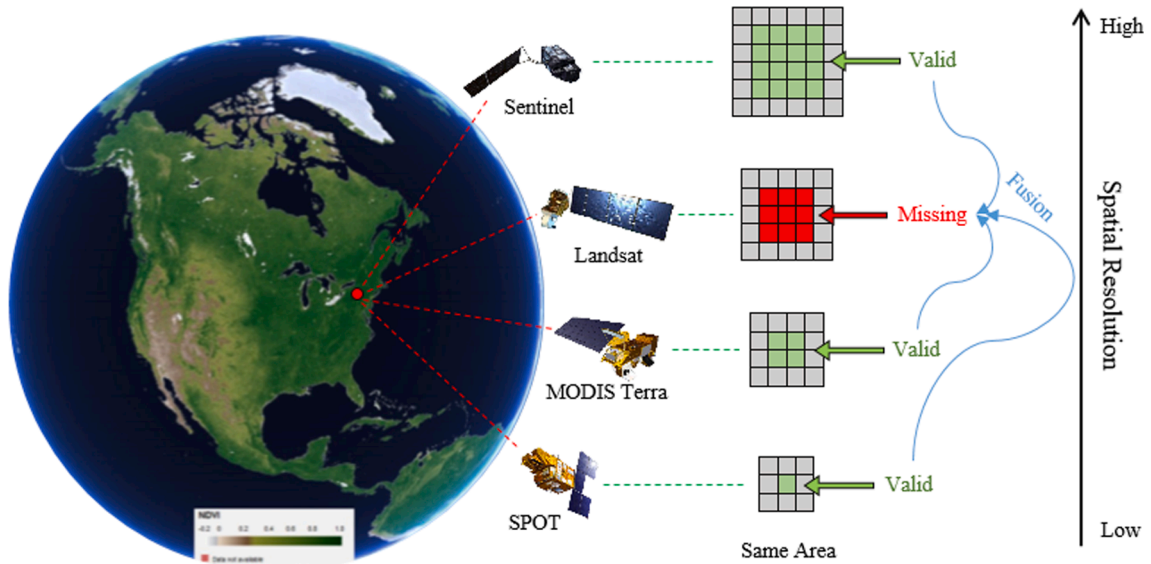
$$\alpha_t(r_i) = \alpha_b(r_i) + \frac{\sum_{j=i-n}^{i+n} w(r_i, r_j) [\alpha_o(r_j) - \alpha_b(r_j)]}{\sum_{j=i-n}^{i+n} w(r_i, r_j)} \quad (18)$$

where n denotes the size of the influence window, $w(r_i, r_j)$ represents the weighting function dependent on the distance between points r_i and r_j by Eq. (19).

Table 6

CC and RMSE of reconstructed results by different methods.

Indicators	CC					RMSE				
	DA	SG	WT	HANTS	STSG	DA	SG	WT	HANTS	STSG
2016	0.6879	0.7255	0.6944	0.7875	0.9761	0.0521	0.0468	0.0570	0.0408	0.0201
2017	0.6745	0.7174	0.6609	0.7311	0.9453	0.0544	0.0484	0.0649	0.0448	0.0272
2018	0.6375	0.6542	0.6748	0.7512	0.9564	0.0706	0.0636	0.0586	0.0430	0.0241
2019	0.6578	0.6642	0.7137	0.6939	0.9413	0.0697	0.0684	0.0470	0.0569	0.0263
2020	0.7225	0.6843	0.6242	0.7168	0.9252	0.0421	0.0518	0.0725	0.0458	0.0314

**Fig. 8.** Multi-source fusion.

$$w(r_i, r_j) = \max\left(0, \frac{R^2 - d_{ij}^2}{R^2 + d_{ij}^2}\right) \quad (19)$$

where R is a predefined radius of influence, d_{ij}^2 means the distance between points r_i and r_j .

TSF is based on a hypothesis that the phenological properties of same vegetation are quite stable over years compared to spatial variations between different pixels (Fang et al., 2007). This method performs well in restoring the spatio-temporal distribution of seasonal trends (Fang et al., 2008). It has effectively improved the data quality in both temporal and spatial aspects.

5.2. Search and fill algorithm with moving offset method

SFA-MOM is a kind of prefilling strategy to reduce data gaps prior to HANTS. It supposes that the same type of vegetation tends to obey a reference curve ($NDVI_{ref}$) of similar phenology (Padhee and Dutta 2019), which is generated by a recursive search and fill algorithm (SFA) using valid NDVI data and corresponding land use/land cover (LULC) data (Friedl et al., 2002). The detailed procedure is shown as Fig. F1 in the Appendix.

The recursive SFA is based on an assumption that vegetation in same ecoregion has similar phenology (Yang et al., 2017). In the circumstances, SFA reconstructs $NDVI_{ref}$ affected with null data using the most similar valid time series of same LULC in a moving window by linear fitting with Eq. (20).

$$RNDVI_{ref,k}^{i,j} = m \times NDVI_{ref,k}^{x,y} + n \quad (20)$$

where (i,j) represents the spatial location of target pixel to be restored, (x,y) represents the spatial location of valid reference pixel in the

window, k denotes the temporal location of target pixel, m and n denote the slope and the intercept of the linear equation between the NDVI time series of target and reference pixel.

In this method, pre-filled NDVI time series are generated by moving offset method (MOM) utilizing original NDVI and smoothed reference NDVI ($SNDVI_{ref}$) as shown in Fig. G1 of the Appendix. MOM is based on two assumptions: 1) the variation of NDVI time series tends to be smooth and continuous (Julien and Sobrino 2010); 2) this kind of change should be nonlinear (Verma and Dutta 2013). Firstly, the offset between valid points in original NDVI time series and its corresponding points in $SNDVI_{ref}$ at the extreme of missing span are interpolated linearly. Then the missing points in original NDVI curve are generated by linear relation with the points in $SNDVI_{ref}$.

As mentioned earlier in Section 2.2, HANTS is inclined to underestimate the NDVI values if there are long-term data gaps. In order to solve this problem, users can adjust the parameter combination or increase data availability. However, the former approach is experience-based and may lead to the loss of valid information. By contrast, the second one is more reliable. SFA-MOM prefills the contaminated time series to increase data availability based on the spatial similarity of same vegetation. This method is proved to be more robust compared with traditional spatial-based or temporal-based methods. Nevertheless, there are some restricted conditions for this method, including the frequency of LULC changes and the accuracy of LULC data, due to the reference NDVI time series are generated based on LULC data.

5.3. Spatio-temporal Savitzky-Golay method

STSG assumes that there are discontinuous clouds in space and uses adjacent pixels to reduce the noise of the target pixels in a given year (Cao et al., 2018). The goals of this method are to restore the continuous

data loss and raise low NDVI values based on

the phenology of vegetation. It is implemented by three steps as shown in Fig. H1 of the Appendix.

For STSG, the initial NDVI time series of target pixels are estimated by the similar neighborhood pixels. The prior knowledge about the growth trajectory is provided by initial NDVI, which can deal with continuous gaps in the original data. In addition, the similar pixels are identified by two criteria: 1) a similar pixel and the corresponding target pixel share a similar land cover type; 2) they have parallel responses to various environmental variations and anthropogenic factors.

Unlike the traditional SG filter (Chen et al., 2004), the weight for each pixel in this iterative SG filter is determined by the Euclidean distance between the original and initial NDVI time series by Eqs. (21) and (22).

$$d_i^j = \begin{cases} \max(0, NDVI_{ori}^{ij} - NDVI_{initial}^{ij}), & RI = 0 \\ NDVI_{ori}^{ij} - NDVI_{initial}^{ij}, & RI = 1 \\ \text{median}\left(\bigcup_{i=1}^{23} d_i^j, RI = 0, 1\right), & RI = -1, 2, 3 \end{cases} \quad (21)$$

where d_i^j represents the Euclidean distance of the i -th day of years (DOY) in the j -th year, $NDVI_{ori}^{ij}$ and $NDVI_{initial}^{ij}$ denote the original and initial time series respectively.

$$W_i^j = \frac{d_i^j - \min(\bigcup_{i=1}^{23} d_i^j)}{\max(\bigcup_{i=1}^{23} d_i^j) - \min(\bigcup_{i=1}^{23} d_i^j)} \quad (22)$$

STSG has considered not only the spatio-temporal relationship, but also the similarity of growth trajectory and responses to environmental variations and anthropogenic factors between different pixels. It can effectively deal with continuous data deficiency and has good performance when processing local low values during harvest. However, STSG is not suitable for the data with spatially continuous clouds because initial NDVI time series can't be generated in this case. In other words, STSG may have poor performance on high spatial resolution data, because they are usually temporally discontinuous due to the low temporal resolution (Cao et al., 2018).

5.4. Spatio-temporal tensor completion method

For all the aforesaid methods, the spatial correlation and temporal continuity of long time series are not being completely utilized. In ST-tensor, the multi-dimensional spatio-temporal information in spatial domain, temporal variations and periodic features are deeply extracted using low-rank adaptive tensor completion, which is applied in NDVI restoration for the first time (Chu et al., 2021). In this method, the multi-year NDVI time series are considered as a tensor generated by the high-order generalization of matrices (Kolda and Bader 2009). The correlations of different domains can be obtained by the tensor's rank with Eq. (23).

$$\min \sum_{n=1}^3 w_n \text{rank}(X_{(n)}) \quad \text{s.t. } X_{\Omega} = Y_{\Omega} \quad (23)$$

where w_n represents the weight of $X_{(n)}$ and satisfies $\sum_{n=1}^3 w_n = 1$, X_{Ω} and Y_{Ω} denote the reliability indices (RI) of tensor X and Y respectively. The weight w_n can be determined by the low-rank degree of the tensor adaptively using singular value decomposition (SVD) (Zha 1991). In addition, an iterative trend filter is used to eliminate the residual noise.

Unlike other hybrid methods, ST-tensor fully considers the spatial, temporal and periodic information of NDVI time series. The spatio-temporal continuous gaps are well settled and the vegetation growth trajectories are faultlessly reserved, which overcome the limitation of STSG. Although ST-tensor has fairly good performance, the uncertainties may be introduced due to the low accuracy of RI data.

In summary, both spatial-based methods and temporal-based methods, which are based on the information of single domain, have benefits as well as drawbacks. In other words, these methods may have good performance under some circumstances but not in others. In order to unite the strengths of different methods and avoid their weakness, hybrid methods using spatio-temporal correlation are proposed. Compared with other methods, hybrid methods can effectively tackle long-term data gaps and preserve the valid low values by utilizing the information in spatial and temporal domains better.

6. Evaluation techniques

The evaluation technique is an indispensable part of NDVI time series reconstruction. In general, in-situ observations are the best benchmark to validate the quality of reconstructed products in real experiments. However, this kind of truthful data are tough to acquire due to the restriction of temporal-spatial resolution or the absence of data sites (Zhou et al., 2015). Hence, the multiyear-synthetic reference NDVI time series, regarded as noise-free data, can be the alternative of ground-based measurements for evaluation. In recent years, they are used extensively in simulated experiments (Atkinson et al., 2012; Hird and McDermid 2009; Julien and Sobrino 2010, 2019; Kandasamy et al., 2013; Michishita et al., 2014; Rahman et al., 2016). They are generally divided into two categories: qualitative assessment and quantitative evaluation.

6.1. Qualitative assessment

Qualitative assessment can be defined as a kind of subjective evaluation by inspecting visually. It is simple, intuitive and easy to capture the changes in NDVI time series. There are mainly two forms of qualitative assessment:

- (1) to observe the time series curve of a single pixel and check whether its changes follow the phenology of this type of vegetation (Jin and Xu 2013; Liu et al., 2017).
- (2) to observe the reconstructed images and the performance is assessed by the clarity of the ground features, the residual degree of noise and the reconstruction track (Cao et al., 2018; Chen et al., 2021).

6.2. Quantitative evaluation

Quantitative evaluation is more convincing by avoiding the differences in subjective assessments. Nevertheless, due to the restriction of spatio-temporal resolution, ground observation is hard to obtain, which is the most immediate data to verify quantitative evaluation. Hence, many studies generate reference NDVI time series from the average of multi-year data to substitute the ground truth. Random noise will be added to reference NDVI time series to test the performance by evaluation indicators (Xu et al., 2015; Zhou et al., 2016a), mainly including RMSE, correlation coefficient (CC), mean absolute error (MAE), universal image quality index (UIQI), structural similarity (SSIM), peak signal to noise ratio (PSNR), etc. Table 6 shows CC and RMSE of reconstructed results by DA, SG, WT, HANTS and STSG. As a result, STSG shows the best performance with the maximum CC and minimum RMSE. According to the simulated experiments, hybrid methods are much better and more robust compared with other techniques.

Although the reconstruction methods are measured by qualitative assessment and quantitative evaluation, there is still no consensus on which one is better. Since the performance of most methods is influenced by the noise distribution (Zhou et al., 2016a), the evaluation results vary. For example, Hird and McDermid, 2009 intensively compared AG, DL, SG, 4523H, MVI and ARMD3-ARMA5, indicating that all the six methods behave good ability in the front ranges of the Rocky Mountains and the performances of AG and DL are better than ARMD3-ARMA5's.

Atkinson et al. (2012) evaluated the results of four methods in India, finding that Whittaker is the best, and AG and DL are not suitable in areas with multiple-growth-season vegetation. On the whole, combining qualitative assessment with quantitative evaluation is able to estimate the curve smoothing and noise alleviating capacity of different methods. However, for the various applications of reconstruction data, this kind of evaluation strategy is insufficient. To remedy this deficiency, vegetation growth trajectory, protection of key point, noise resistance and curve stability were considered as assessment indicators for different applications (Liu et al., 2017). Hence, this kind of task-oriented strategy can provide traditional evaluation techniques with valid reference.

7. Development trends

Although lots of techniques have been proposed to reconstruct NDVI time series, there are still many aspects for improvement. We think the future development trends include: 1) how to effectively utilize spatial and temporal information together (spatio-temporal combination); 2) how to utilize multi-source data (multi-source fusion); 3) how to apply more machine learning methods (machine learning adoption); 4) how to embed task-driven idea into reconstruction (task-driven strategy); 5) how to improve the performance and efficiency simultaneously (good performance and high efficiency).

(1) Spatio-temporal Combination

Up to now, most studies singly consider spatial or temporal correlation of NDVI time series. These temporal-based methods can effectively fit the growth curve of vegetation to some extent. However, temporal correlation may be invalid when there are long-term data gaps. It is imperative to reduce this kind of data gaps utilizing spatial information. Indeed, some researches have tried spatio-temporal combination methods, like the hybrid methods as stated previously and other potential solutions, such as fit-and-fit (FF) approach for Landsat surface reflectance Analysis Ready Data (ARD) gap-filling (Yan and Roy 2020), adaptive spatio-temporal weighted method (ASTWM) for MODIS snow cover products reconstruction (Li et al., 2017) and Generalised Additive Model (GAM)-geostatistical space-time model for MODIS enhanced vegetation index (EVI) denoising (Poggio et al., 2012). Although these methods are designed for other data, they may also be applied to reconstruct NDVI time series with appropriate adjustment. Hence, we believe spatio-temporal combination will be one of the development trends in NDVI time series reconstruction. The key points of these methods are to effectively utilize spatio-temporal information together.

(2) Multi-source Fusion

For all the methods mentioned in this review, there is a common problem: the limitation of single spatial or temporal resolution. This will lead to low quality results when spatial or temporal information is deficient. However, multi-source datasets of different resolutions can provide additional information to make up for the deficiency of single-source data. Generally speaking, if some sensor suffers from cloud, while the data of other sensors may be cloud-free in the same area. Hence, the contaminated data can be restored by spatio-temporal fusion as shown in Fig. 8. This idea has already been applied in some other analogous research field, such as kernel-based nonparametric data assimilation for land surface temperature (LST) reconstruction with Global land skin temperature (Ts) from Global Land Data Assimilation System (GLDAS) (Padhee and Dutta 2020), the fusion approach for MODIS leaf area index (LAI) gap-filling using SPOT VEGETATION and MODIS reflectance products (Verger et al., 2011). Also, some classical spatio-temporal fusion models generating high spatio-temporal resolution data could be potential techniques, like Spatiotemporal fusion method to Simultaneously generate Full-length normalized difference vegetation Index Time series (SSFIT) (Qiu et al., 2021), Enhanced

Spatial and Temporal Adaptive Reflectance Fusion Model (ESTARFM) (Zhu et al., 2010) and Flexible Spatiotemporal Data Fusion (FSDAF) (Zhu et al., 2016). In our view, the fusion of multi-source data is also a promising direction.

(3) Machine Learning Adoption

Compared with natural images, remote sensing images have more complicated patterns and information, which may reduce the availability of traditional methods. Different from conventional techniques, machine learning has strong ability in feature expression which can accurately capture the potential associations between environment variables (Yuan et al., 2020). On the one hand, some researchers have tried deep learning networks to reconstruct remote sensing data, such as spatial-temporal-spectral deep convolutional neural network (STS-CNN) for missing information restoration (Zhang et al., 2018b), global-local deep convolutional neural network (DCNN) for thick cloud removal (Zhang et al., 2020), as well as long short-term memory (LSTM) network for MODIS LST image reconstruction (Arslan and Sekertekin 2019). These deep learning models are highly portable for similar studies. On the other hand, NDVI time series data have been accumulated for a long time, which can provide the learning methods with sufficient training samples. Hence, we think more machine learning techniques will be applied in the reconstruction of NDVI time series.

(4) Task-driven Strategy

NDVI time series reconstruction is mainly to provide high-quality products for the subsequent applications. In this review, all the mentioned techniques are method-driven or data-driven. However, different tasks like vegetation classification and phenology extraction have different requirements. Hence, the restoration algorithms are supposed to be capable of generating appropriate products for different tasks. The task-driven strategy algorithms have already been mentioned in some previous works (Latif et al., 2008; Li et al., 2016; Weiss et al., 2014). In brief, more subdivided task-driven reconstruction methods will be proposed according to different situations in the future.

(5) Good Performance and High Efficiency

Although dozens of techniques have been proposed, some may have good performance with low efficiency, while others may possess high efficiency but bad performance. In the ideal case, the reconstruction algorithm is supposed to run as quickly as possible on condition that the performance is good enough. We believe that the balance between the performance and efficiency of algorithms is another research focus, as well as the ultimate aim of this field.

8. Conclusions

Due to the influence of multiple interference, NDVI time series derived from satellite are generally contaminated. Hence, accurate reconstruction of noise gaps is quite important. Over the past few decades, plenty of algorithms have been proposed to smooth NDVI time series. According to the principle of reconstruction, these techniques are classified into three categories: 1) temporal-based methods; 2) frequency-based methods and 3) hybrid methods. These methods are based on different variation characters, specific assumptions or reality of vegetation phenology. Based on temporal variation continuity, the results of temporal-based methods meet the phenological regularity of vegetation well with fairly good performance unless there are long-term data gaps. Frequency-based methods reconstruct NDVI time series by transforming them into frequency domain. As for hybrid methods, the information of spatial correlation and temporal continuity is considered to break through the limitation of single feature. In this review, we introduced the detailed procedure, benefits and drawbacks of each

method, the evaluation on reconstruction performance as well as the development trends of this research direction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Figs. A1-H1.

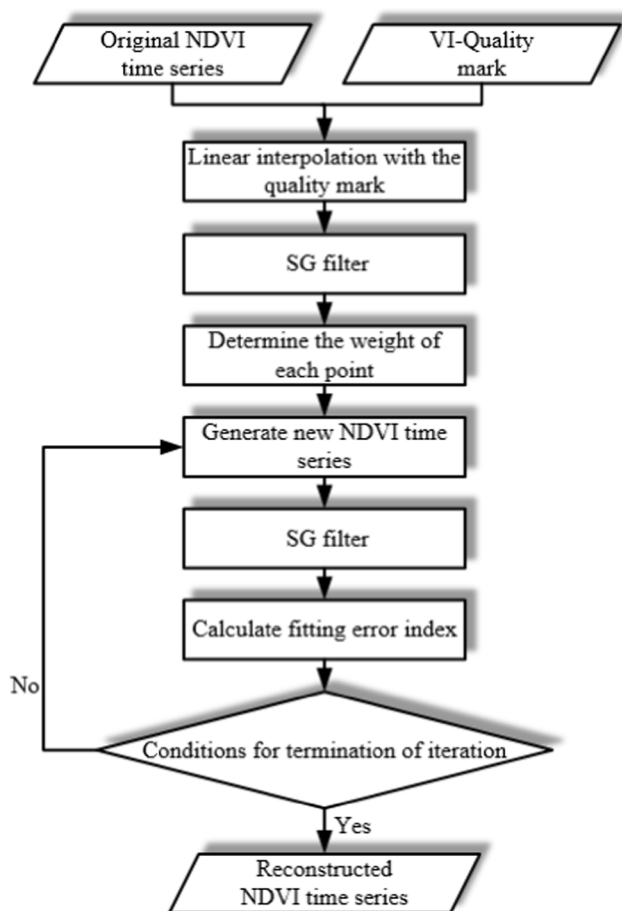


Fig. A1. The procedure of SG filter (Chen et al., 2004).

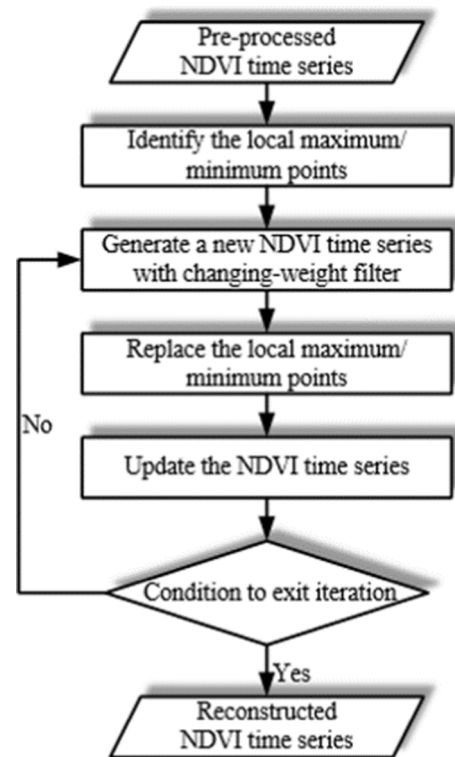


Fig. B1. The procedure of CW filter (Zhu et al., 2012).

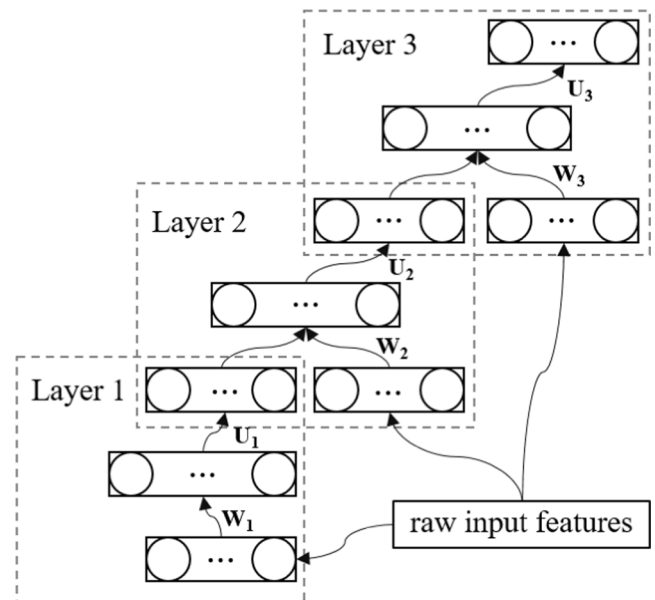


Fig. C1. The architecture of DSN (Nie et al., 2014).

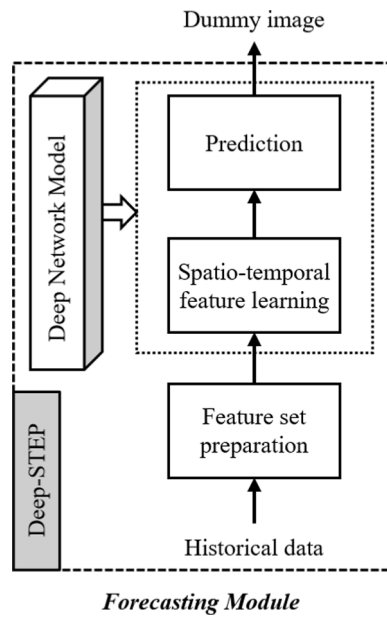


Fig. D1. The forecasting module of Deep-STEP FE (Das and Ghosh 2017).

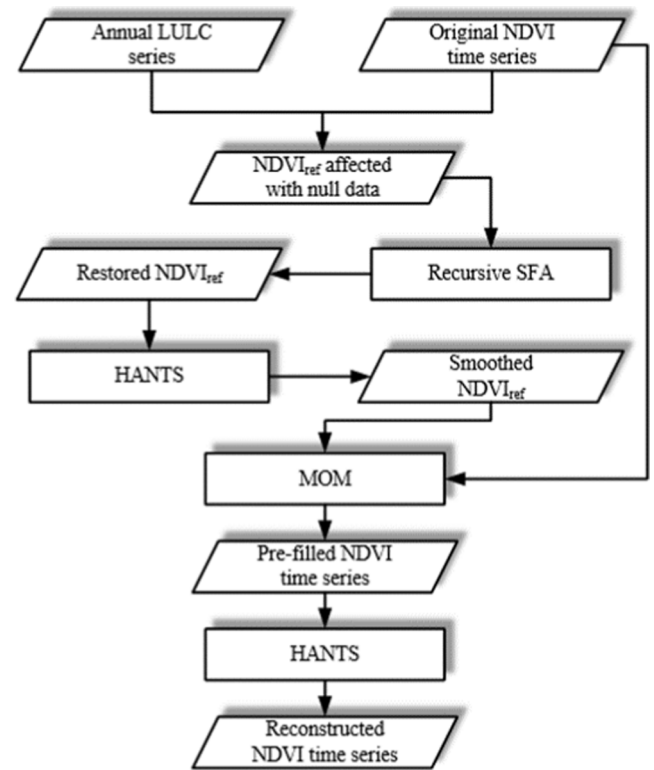


Fig. F1. The procedures of SFA-MOM (Padhee and Dutta 2019).

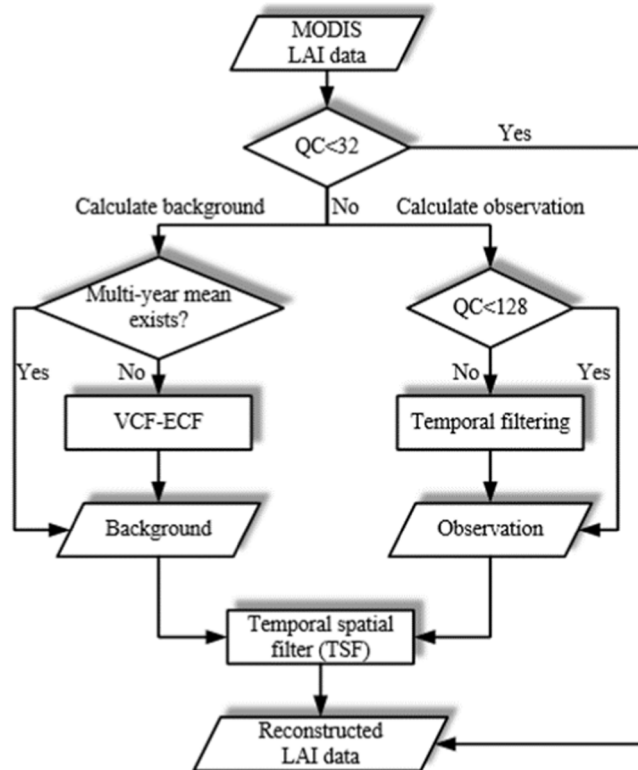


Fig. E1. The procedure of TSF (Fang et al., 2007).

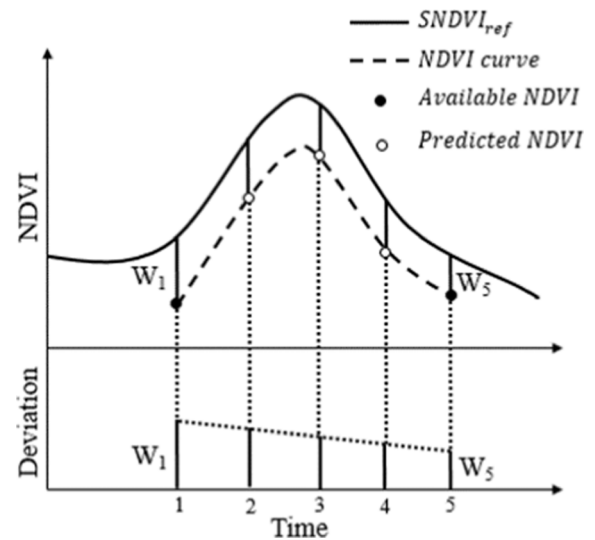


Fig. G1. The conceptual demonstration of MOM (Padhee and Dutta 2019).

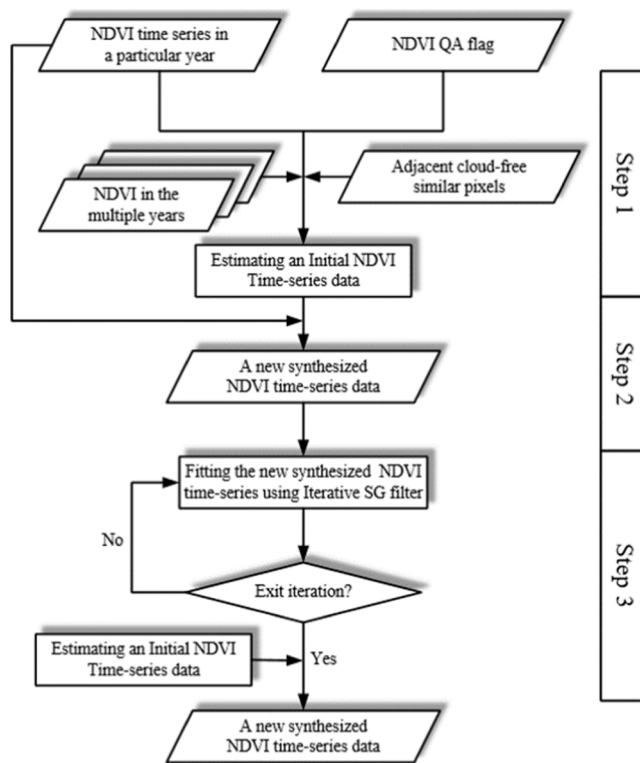


Fig. H1. The procedures of STSG (Cao et al., 2018).

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